

$$= 2500 \cdot 10 \cdot 400^2 = 4000000000 = 4 \text{ TH}$$

$$\varphi = \arctg \frac{400^2}{500 \cdot 10} = \arctg \frac{160000}{5000} =$$

$$\approx \arctg 32 \approx 88,2^\circ$$

$$\text{Given: } \varphi = 88,2^\circ, F = 4 \text{ TH}$$

Distance $\frac{1}{10}$ of λ

on $10, 10$

$$\lambda \approx 7,43$$

Given:

$$L = 0,5 \text{ m}$$

$$m = 0,36 \text{ kg}$$

$$d = \frac{1}{12} \text{ m}$$

$J = ?$

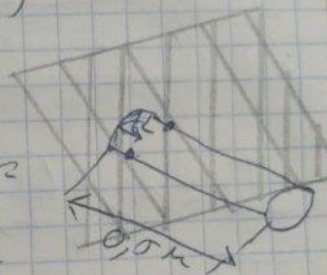
Then:

$$J_{cm} = \frac{1}{12} mL^2 =$$

$$= \frac{0,36 \cdot 0,25}{12} =$$

$$= 0,0025 \frac{\text{kg} \cdot \text{m}^2}{12}$$

$\approx 0,0025 \frac{\text{kg} \cdot \text{m}^2}{12}$



$$J = J_{cm} + md^2; J = 0,0025 + 0,36 \cdot \frac{1}{144} \approx$$

$$\approx 0,01 \frac{\text{kg} \cdot \text{m}^2}{12}$$

$$\text{Answer: } J \approx 0,0070 \frac{\text{kg} \cdot \text{m}^2}{12}$$

Dano:

$$m = 0,1 \text{ kg}$$

$$m_1 = 0,4 \text{ kg}$$

$$m_2 = 0,3 \text{ kg}$$

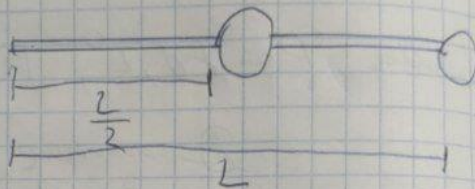
$$L = 0,8 \text{ m}$$

$$l_1 = \frac{L}{2}, l_2 = L$$

J-?

~ 2. (4.9)

Terminale:



$$J_{cm} = \frac{1}{3} m L^2 = \frac{0,1 \cdot 0,64}{3} \approx$$

$$\approx 0,0213 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

$$J_1 = m_1 \left(\frac{L}{2} \right)^2 \approx 0,064 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

$$J_2 = m_2 L^2 \approx 0,196 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

$$J = J_{cm} + J_1 + J_2 \approx 0,0213 + 0,064 + 0,196 =$$

$$\approx 0,2773 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} \approx 0,3 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

Odpowiedź: $0,3 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$

Dano:

$$m_1 = 0,$$

$$m_2 = 0,$$

$$a = 0,$$

a-?

Due

Due

$$J_{\beta} =$$

$$\Rightarrow$$

$$h.k.$$

13. (3.25)

Решение!

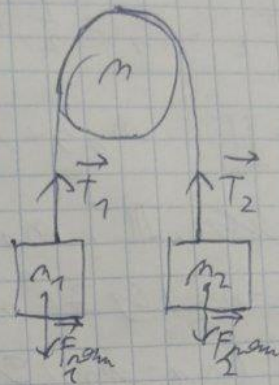
Дано:

$$m_1 = 0,7 \text{ кг}$$

$$m_2 = 0,71 \text{ кг}$$

$$R = 0,4 \text{ м}$$

$$a = ?$$



Для I груза (II закон Н.):
 $m_1 g - T_1 = m_1 a$

Для II груза:
 $T_2 - m_2 g = m_2 a$

$$J_{\beta} = R(T_2 - T_1); \quad a = \beta R \Rightarrow \beta = \frac{a}{R}$$

$$\Rightarrow R(T_2 - T_1) = J \frac{a}{R}; \quad T_2 - T_1 = J \frac{a}{R^2}$$

$$\text{так } J = \frac{1}{2} m R^2, \text{ то}$$

$$T_2 - T_1 = \frac{\frac{1}{2} m R^2 \cdot a}{R^2} = \frac{1}{2} m a$$

$$J = mR^2$$

Do II Zakony: $m_1 g - T_1 = m_1 a$
 $T_2 - m_2 g = m_2 a$

Można: $T_2 - T_1 = \frac{mR^2 a}{R^2} = ma$

Suma równań y - u - u :

$$\begin{cases} m_1 g - T_1 = m_1 a \\ T_2 - m_2 g = m_2 a \end{cases} \Rightarrow a = \frac{(m_2 - m_1)g}{m_1 + m_2 + m} =$$

$$T_2 - T_1 = ma$$

$$= \frac{0,2 \cdot 10}{1} = 2 \frac{m}{s^2} \Rightarrow$$

$$\Rightarrow T_1 = m_1 g - m_1 a = 0,3 \cdot 10 - 0,3 \cdot 2 = 2,6 \text{ N}$$

$$\Rightarrow T_2 = T_1 + m a = 2,6 + 0,2 \cdot 2 = 3 \text{ N}$$

Odpowiedź: $T_1 = 2,6 \text{ N}$, $T_2 = 3 \text{ N}$
 $\sim 5. (4.2.)$

Dane:

$$J = 63,6 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

$$\omega_0 = 31,4 \frac{\text{rad}}{\text{s}}$$

$$t = 20 \text{ s}$$

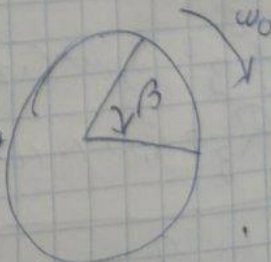
$M = ?$

Terminy:

$$\beta = \frac{\Delta \omega}{t}$$

$$\Delta \omega = \omega_0 - 0 = 31,4 \frac{\text{rad}}{\text{s}}$$

$$\beta = \frac{31,4}{20} = 1,57 \frac{\text{rad}}{\text{s}^2}$$



$$M = J\beta = 63,6 \cdot 1,52 \approx 100 \text{ Н}\cdot\text{м}$$

Ответ: $100 \text{ Н}\cdot\text{м}$